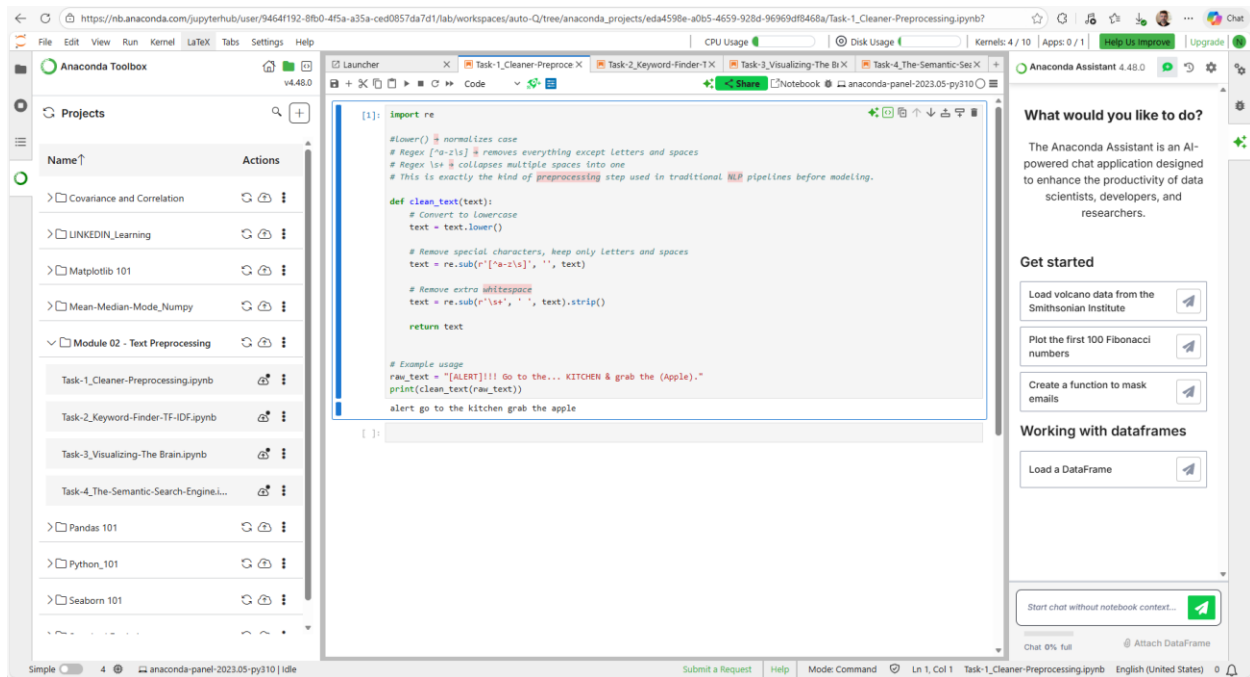


Module 02 - Text Preprocessing, TF-IDF, Word Embeddings, Dimensionality Reduction & Semantic Search Engine - Cosine Similarity



```
[1]: import re

#lower() & normalizes case
# Regex [a-z]+ & removes everything except letters and spaces
# Regex \s+ & collapses multiple spaces into one
# This is exactly the kind of preprocessing step used in traditional ML pipelines before modeling.

def clean_text(text):
    # Convert to lowercase
    text = text.lower()

    # Remove special characters, keep only letters and spaces
    text = re.sub(r"[^a-zA-Z]", '', text)

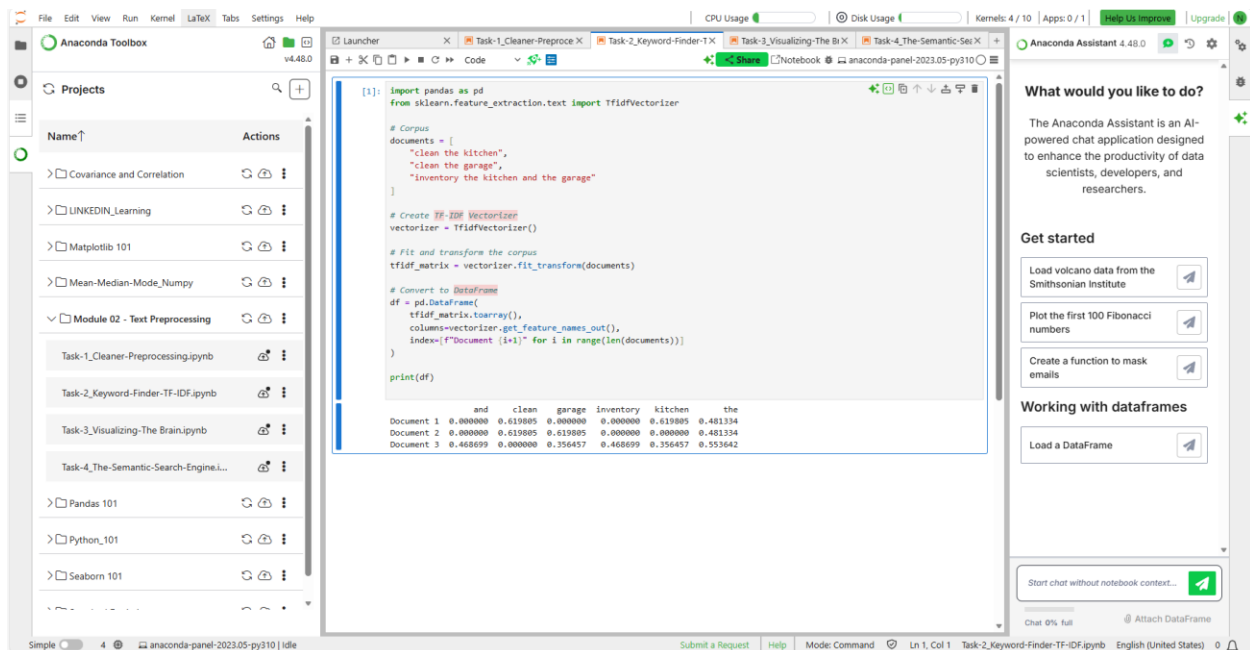
    # Remove extra whitespace
    text = re.sub(r"\s+", ' ', text).strip()

    return text

# Example usage
raw_text = "I!!! go to the... KITCHEN & grab the (Apple)."
print(clean_text(raw_text))

alert go to the kitchen grab the apple

[ ]:
```



```
[1]: import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer

# Corpus
documents = [
    "clean the kitchen",
    "clean the garage",
    "inventory the kitchen and the garage"
]

# Create TF-IDF Vectorizer
vectorizer = TfidfVectorizer()

# Fit and transform the corpus
tfidf_matrix = vectorizer.fit_transform(documents)

# Convert to DataFrame
df = pd.DataFrame(
    tfidf_matrix.toarray(),
    columns=vectorizer.get_feature_names_out(),
    index=[f"Document {i+1}" for i in range(len(documents))]
)

print(df)
```

	and	clean	garage	inventory	kitchen	the
Document 1	0.000000	0.619805	0.000000	0.000000	0.619805	0.481334
Document 2	0.000000	0.619805	0.619805	0.000000	0.000000	0.481334
Document 3	0.468699	0.000000	0.356457	0.468699	0.356457	0.553642

